CPE 301: Embedded Systems Design Final Project Part 3 Assignment

Individual / Group members' names: Abagail Sheats and Nathaly Naranjo

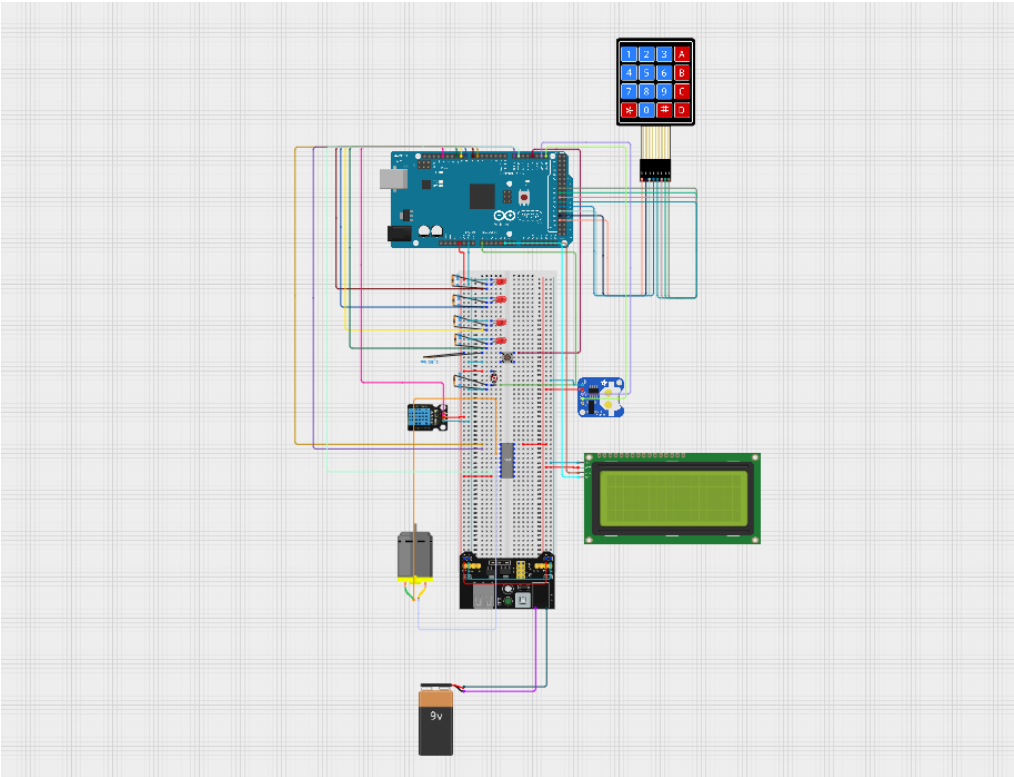
1. Project Motivation/Introduction

Our system is designed to function similarly to an air conditioning unit by automatically adjusting fan speed to maintain a cooler and more comfortable environment. Cooling systems like this are important in everyday life, especially in warmer climates, residential homes, commercial buildings, and outdoor settings. An embedded system makes this possible by processing sensor data, such as temperature and light levels, to control the fan efficiently; for example, the fan speed is controlled by specified temperatures. Overall, the project focuses on improving comfort while integrating energy-saving features through automated control.

2. Proposed System Overview

Our system will read the temperature and detect the brightness of the environment in which it is placed. Our system will only collect this data once the user turns it on with the push of a button. Then, utilizing the data, the fan will turn on and adjust the speed accordingly. If the photoresistor detects a dark environment in its idle state, then the system will not turn on until light has been detected. A button on the matrix keypad serves as the reset in case of an error, such as sensor values being out of range.

2.1 System Schematic Diagram



2.2 Component List

Red LED (Error State)
Blue LED (Off State)
Yellow LED (Idle State)
Green LED (Active state)
1 Push Button (On State)
Matrix Keypad (Reset (1) and Off Buttons (0))
DC Motor (Fan)
Motor Driver IC (L293D)
DHT11 Sensor (Temp/Humidity Sensor)
Photoresistor (Reads light)
RTC DS 1307 (Serial Monitor Logging)
LCD (Sensor Values and State)
Power Supply Module
9 Volt Battery Supplier

3. Expected System Behavior (High-Level)

The goal of the system is to reduce high temperatures while also saving power. When our system is turned on through a button, the system will collect analog data of the brightness of the environment and send it to digital data. The temperature will determine the speed at which the fan will spin, while the brightness will determine whether the fan is on at all. When the fan is on, we want the motor to be controlled based on how cold or hot the room is, so the fan speed will be predicted by the temperature. When our system is off, there should be no activity coming

from our system. The times we should see major changes are when our fan detects a temperature below 15.6 degrees Celsius, which will send the system into Error state. Additionally, if the temperature happens to increase or decrease below or above a specific threshold, it will change the fan speed. For example, between the degrees (in Celsius) 15.6 – 26.67, the speed will be low (90), between 26.76-32.22, the speed will be low (150), and finally, from temperatures 32.22 and above, the speed is the highest at 255.

3.1. Module Design

<u>Sensor reading module</u>

- This module will read analog data from a photoresistor and digital data from a temperature and humidity module. It will then send the data over to the control logic, display, and communication module
- The data read from this module will determine what the system performs next, or whether it will act.

<u>Control logic module</u>

- This module will take the data from the sensor reading module and compare it to a threshold to determine which state our system should be in.
- Firstly, it determines whether the room is dark or bright. If dark, the fan will be off; otherwise, active state will be entered.
- If the temperature reads between 15.6-26.67 C then the fan stays on low speed; if it reads between 26.67 – 32.22 then the fan will be at medium speed, otherwise it will be at the max speed.
- If the temp falls below 15.6 then there is an error and ERROR state will be entered

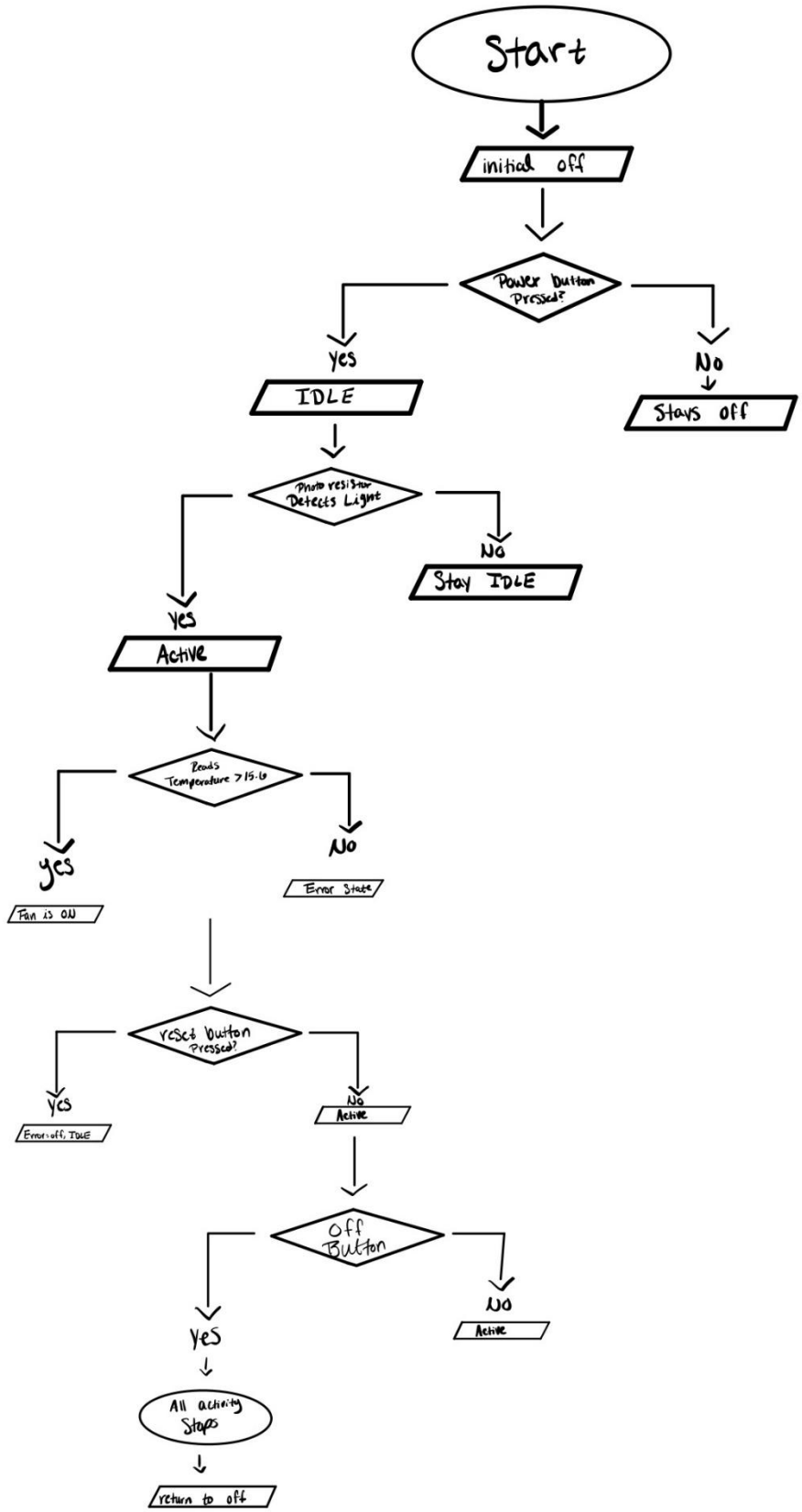
<u>Display module</u>

- This module will display information gathered by the sensor reading module on an LCD. It will display new data every minute, including the temperature and how much light is in the room.

<u>Communication (serial) module</u>

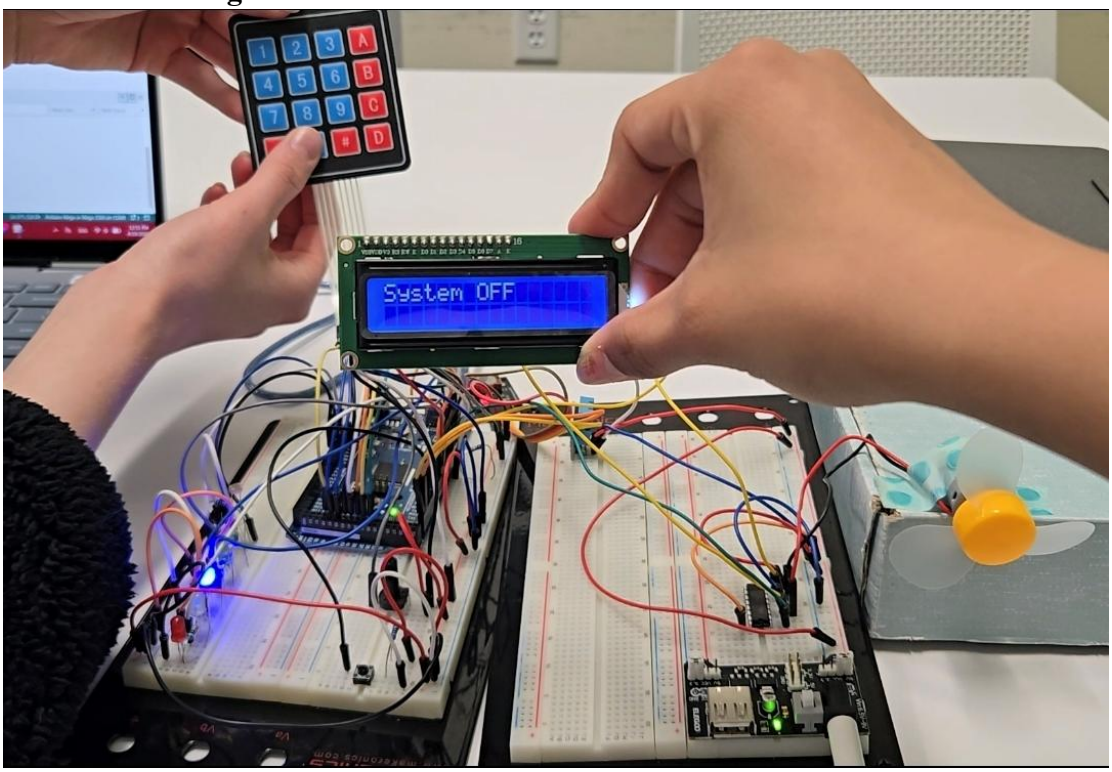
- This module displays information from the control logic module, with time stamps.
- It shows when the system enters a new state, displays any error events, and states when the actuator turns on or off through the Serial Monitor

3.2 Control Logic (State Diagram or Flowchart)


Start
initial off
Power Button Pressed?
yes
IDLE
No
Stays off
Photo resistor Detects Light
No
Stay IDLE
yes
Active
Reads Temperature > 15.6
yes
Fan is ON
No
Error State
reset Button Pressed?
yes
Error off, IDLE
No
Active
Off Button
yes
All activity Stops
return to off
No
Active

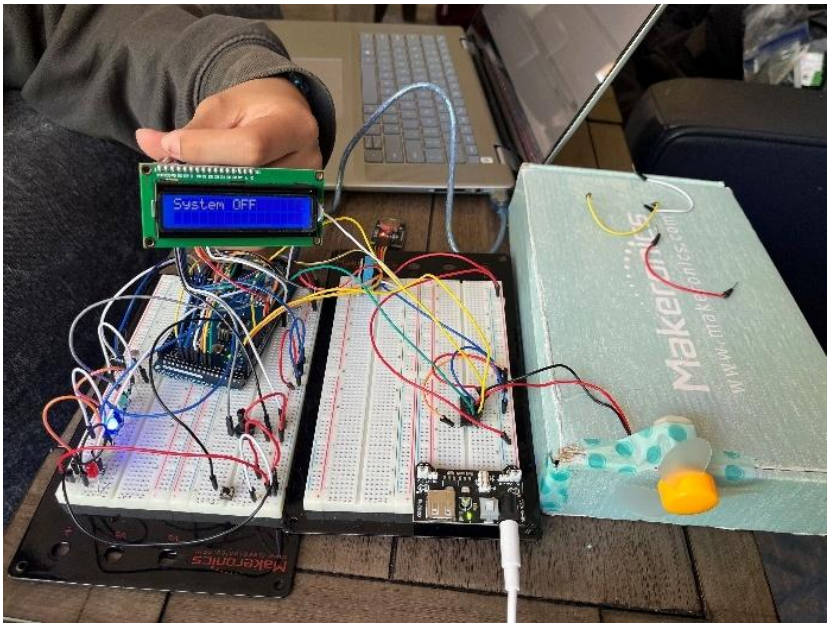
4. Results

Full Circuit Image:

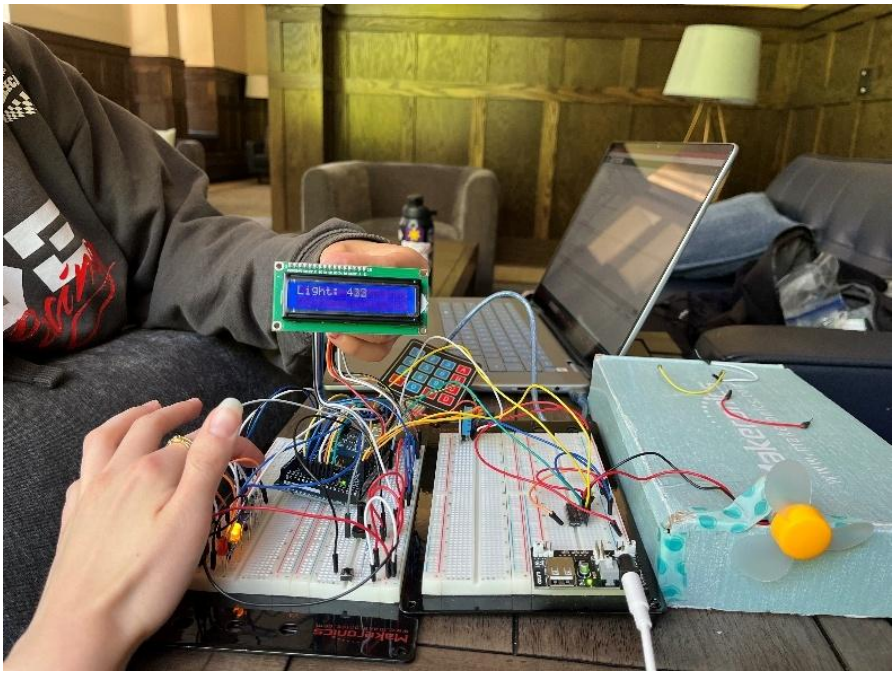


Working Images For Each State:

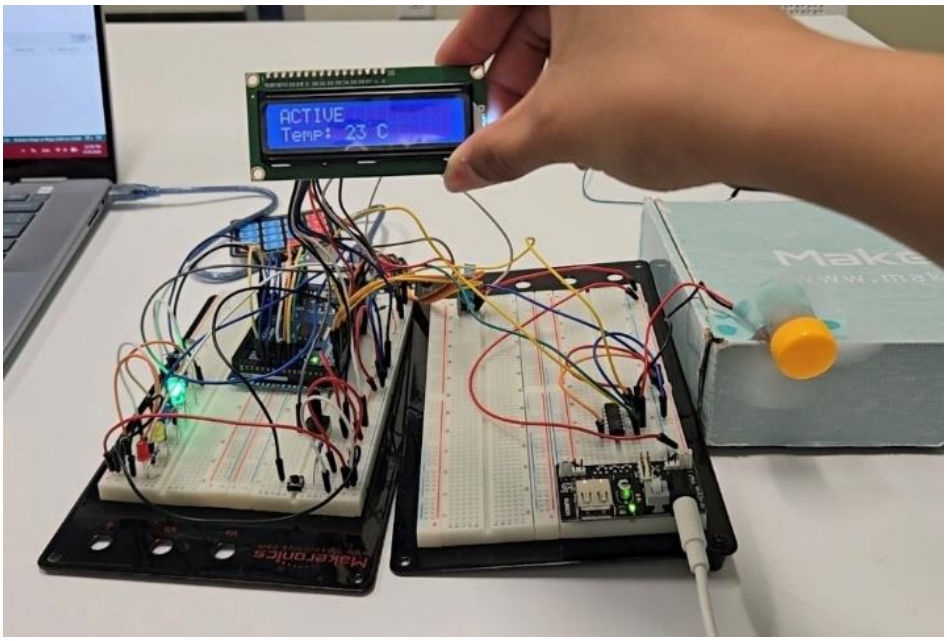
OFF



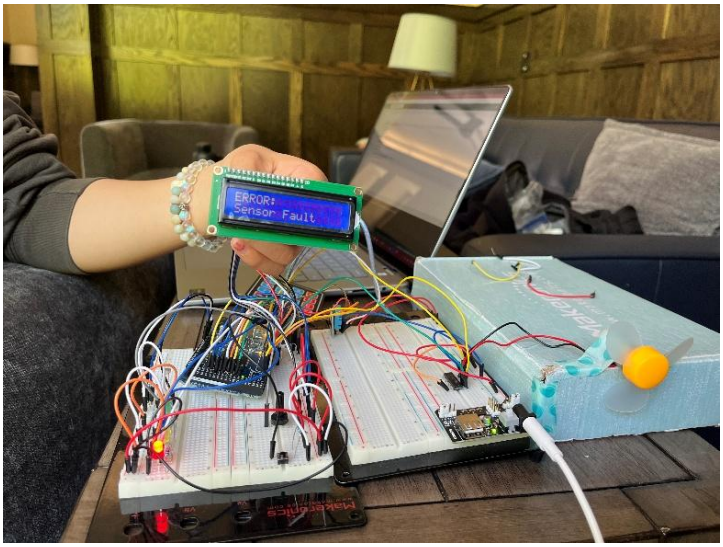
IDLE



ACTIVE



ERROR



Serial Monitor Screenshots:

```
11:36:55.285 -> 11:36:48 IDLE
11:36:56.390 -> 11:36:49 ACTIVE State Entered
11:36:56.424 -> 11:36:49 Fan On
11:37:02.240 -> 11:36:55 System OFF
11:37:03.939 -> 11:36:57 IDLE
11:37:03.972 -> 11:36:57 ACTIVE State Entered
11:37:04.064 -> 11:36:57 Fan On
11:37:36.525 -> 11:37:29 IDLE
11:37:36.525 -> 11:37:29 ACTIVE State Entered
11:37:36.594 -> 11:37:29 ERROR: Sensor Out Of Range
```

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5. Environmental & Energy Impact

- **Energy Efficiency:** Our fan system will be more energy efficient than regular fans due to its adaptability. Our idle state will ensure that, while it is dark in a room, the fan will not run. If it detects light, then active state will begin which means that temperature sensor data will begin being read and the fan speed will change accordingly to ensure that extra energy is being conserved if the environment allows it.
- **Design Safety:** The design of our circuit will be first tested through TinkerCad to play around with the setup without risking electric shocks, burning electric components, or ruining our breadboard. Additionally, we will perform the proper calculations necessary

when determining the proper resistor to use with our components. Thorough testing of the circuit will help limit any issues with the hardware on the physical breadboard.

- **Affordability:** The system will only utilize components from the ATMEGA Starter Kit, but the individual components only include a temperature and humidity module (retails about $5), fan blade with motor (retails about $6.49), a photoresistor (retails about $0.95), resistors and a button (about $2.49). Moreover, this project would only take about $14.93 to make without including the cost for the breadboard. This is affordable considering that most fans cost upwards of $40, and those are not energy efficient such as ours will be.
- **Sustainability:** The design of our fan system is easily repairable in any instance. The circuit will heavily rely on the data it reads rather than an excess of hardware components. Our hardware is inexpensive and easy to replace at home.
- **Accessibility:** The design of our system will be accessible due to the adaptive nature of the fan. Moreover, it does not require much human intervention since the fan will adjust its speed based on the temperature, ensuring it is neither too cold nor hot.

6. Conclusion

The embedded system functioned as anticipated. In the fan project, digital data from a temperature sensor and analog data from a photoresistor were utilized to determine the operational state and speed of the fan blade. However, the project is constrained by the components provided in the starter kit, which includes a small fan blade and a 9V battery as the power supply for the DC motor. Ideally, this project would be improved by simulating it with a larger motor, a bigger fan blade, and a more substantial power supply to create a truly interactive smart fan.